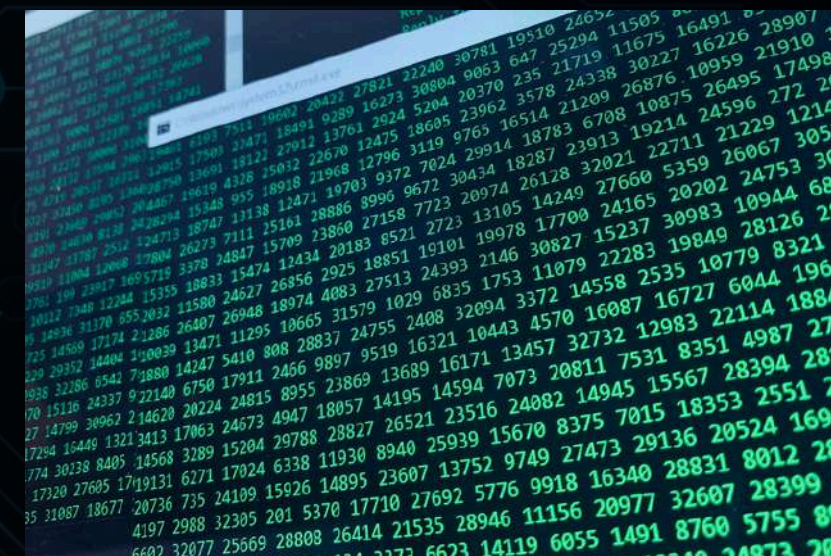
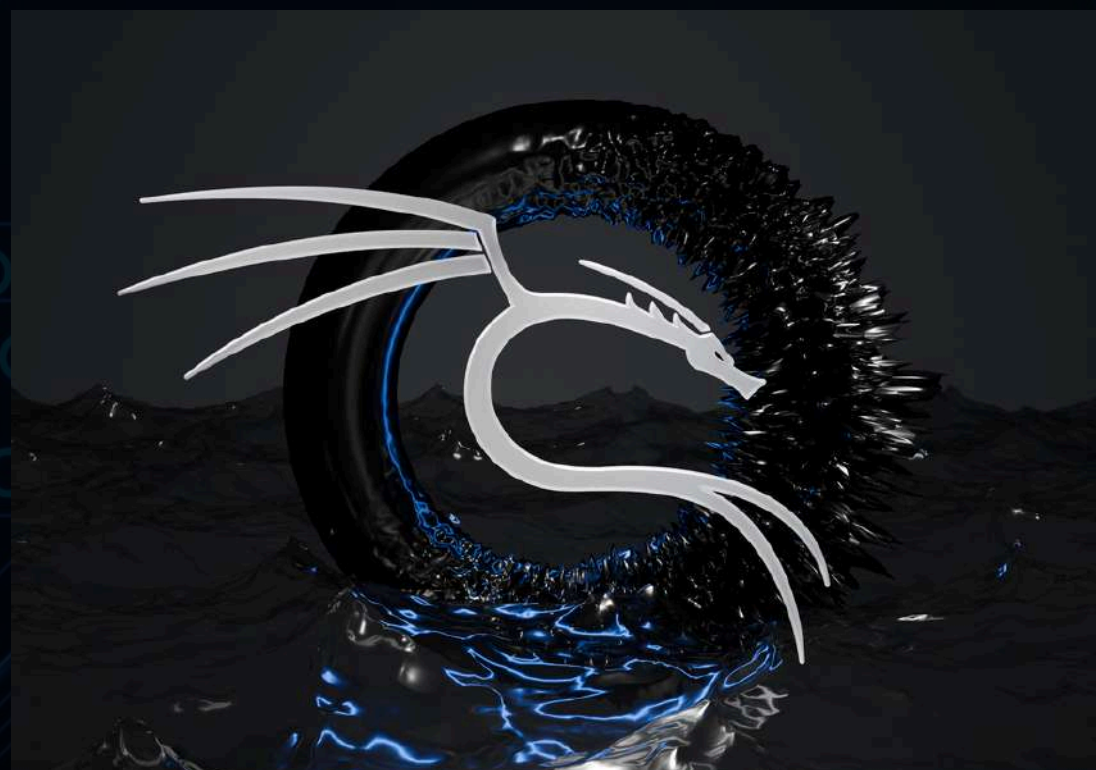


INTRO TO CTF & KALI LINUX

ICECTF WORKSHOP 2.0
10 OCT 2026



REMINDER

- **Don't worry if you don't get it right away.** Learning takes time, and everyone learns at their own pace.
- **Ask questions and learn by doing.** There are no dumb questions, and practice helps things click.
- **Support each other. We all make mistakes, including me, sooo feel free to correct me!**

WHAT WILL BE COVERED TODAY?

GETTING STARTED

- VMware & Kali Linux
- File structure, Shell & basic commands

INTRO TO CTF

- What is a CTF?
- CTF formats & Jeopardy categories

HANDS-ON CTF

- Solve beginner-friendly challenges



VMWARE

VIRTUAL MACHINE

- Allows you to **run multiple operating systems** from one host or laptop
- No need to change current operating system
- Operating system installed within the VM behaves as if installed directly on the hardware
- You can **snapshot VM to save configuration**
- E.g. : VMWare, VirtualBox, Hypervisor



VMWARE: WHAT AND WHY?

- A **virtualization software** that lets you run multiple operating systems (like Linux, Windows, etc.) on a single physical computer.
- Open any malicious file **without changing or damaging your host** computer.
- No need for dual booting, just run Linux and Windows side by side.
- You can save a VM state and easily revert back if something breaks.
- Allocate CPU, memory, and storage to each VM as needed.
- Works on Windows, macOS, and Linux hosts.

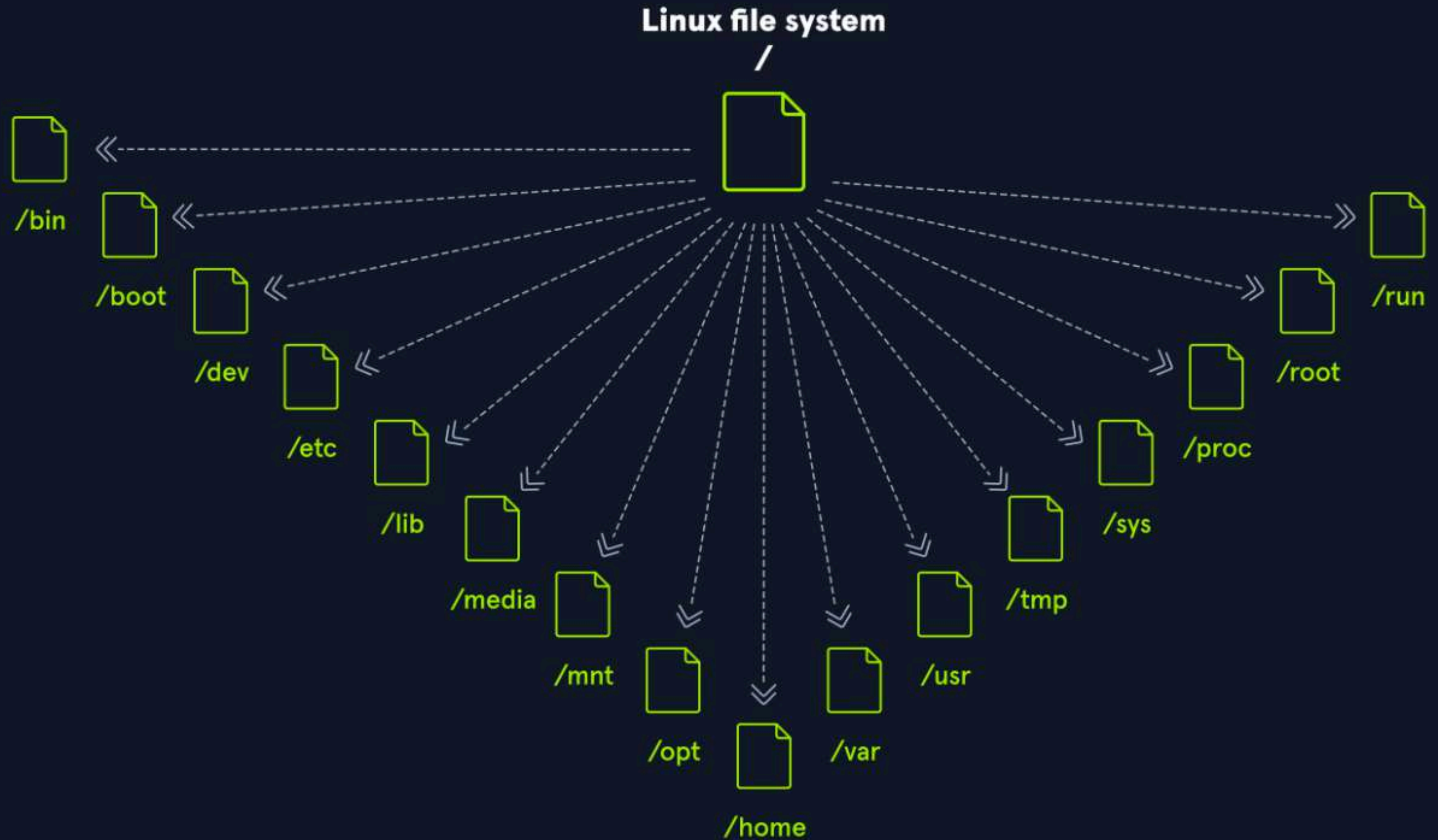


KALI LINUX

KALI LINUX: WHAT AND WHY?

- Kali Linux is a **Linux operating system** designed for **cybersecurity** and **penetration testing**.
- It comes with many **pre-installed security tools** for tasks like network analysis, vulnerability scanning, password testing, digital forensics, and exploitation.
- **All-in-one toolkit:** Tools like Nmap, Wireshark, Burp Suite, and John the Ripper are readily available.
- **Great for CTFs:** Makes it easier to investigate and solve challenges involving web, network, cryptography, forensics, reverse engineering, and exploitation.

FILE SYSTEM HIERARCHY



FILE SYSTEM HIERARCHY

Path	Description
/	The top-level directory is the root filesystem and contains all of the files required to boot the operating system before other filesystems are mounted, as well as the files required to boot the other filesystems. After boot, all of the other filesystems are mounted at standard mount points as subdirectories of the root.
/root	The home directory for the root user.
/tmp	The operating system and many programs use this directory to store temporary files. This directory is generally cleared upon system boot and may be deleted at other times without any warning.
/etc	Local system configuration files. Configuration files for installed applications may be saved here as well.
/home	Each user on the system has a subdirectory here for storage.
/var	This directory contains variable data files such as log files, email in-boxes, web application related files, cron files, and more.



SHELL & BASIC COMMAND

INTRO TO SHELL

- Linux terminal or **command line**
- Provides a text-based input/output (I/O) interface between users and the kernel
- Most commonly used shell in Linux is the Bourne-Again Shell (**BASH**).
- Interact with programs and processes to get information faster



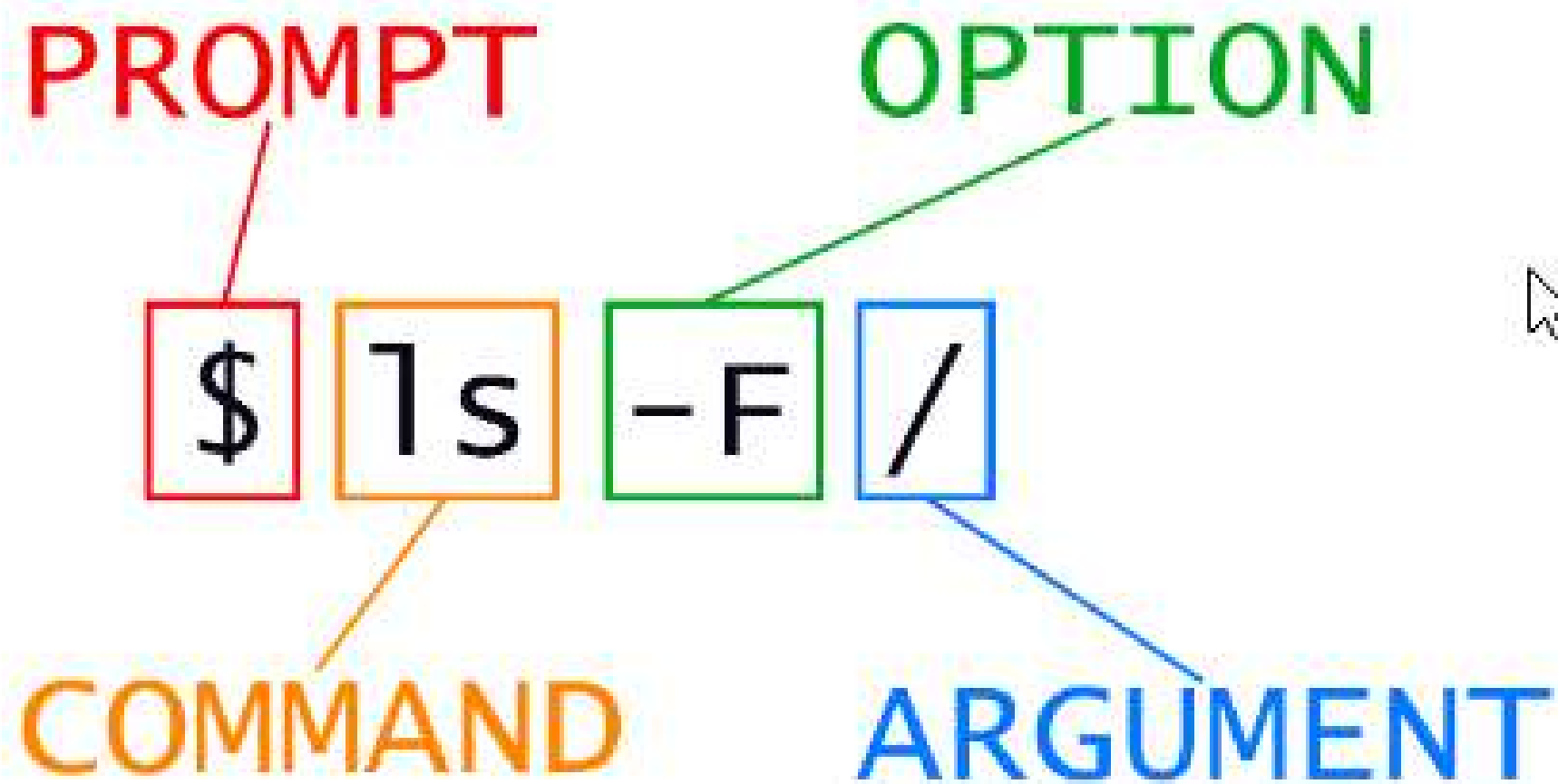
```
File Actions Edit View Help
kali@kali:~$ uname -a
Linux kali 5.9.0-kali1-amd64 #1 SMP Debian 5.9.1-1kali2 (2020-10-29) x86_64 GNU/Linux

kali@kali:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Kali
Description:    Kali GNU/Linux Rolling
Release:        2020.4
Codename:       kali-rolling

kali@kali:~$ echo "www.linuxni.com"
www.linuxni.com

kali@kali:~$ cd www.linuxni.com
kali@kali:~/www.linuxni.com$
```

COMMON COMMAND SYNTAX



example: \$ ls -l /home

PROMPT DESCRIPTION

- Format can look something like this :

```
kali@kali:[/etc]$
```

- The home directory for a user is marked with a tilde <~> and is the default folder when we log in.

```
kali@kali:[~]$
```

- The dollar sign, in this case, stands for a user. As soon as we log in as root(su -), the character changes to a hash <#> and looks like this:

```
root@kali[/etc]#
```

BASIC COMMAND

Command	Description	Example Usage
mkdir	creates a new directory (folder)	mkdir <folder name>
pwd	Print Working Directory. Print path for our current directory	pwd
ls	List all files or folder in directory	ls ls -la
cd	changing directory	cd <folder name> cd .. cd ~
echo	Display a line of text or string on the terminal. It's commonly used for printing messages, debugging scripts, or sending output to files	echo "Hello, World!" echo "This is a test" > file.txt
cat	Shorts for concatenate. Used to display contents of a file	cat filename.txt

BASIC COMMAND - CONT.

Command	Description	Example Usage
sudo	Super User Do. Permit user to run as superuser(root).	sudo apt update sudo vim /etc/hosts
clear	Clear terminal view	clear [CTRL + L]
grep	Searches for a specific pattern of characters in files or output and displays the matching lines	cat filename.txt grep "search_terms"
ifconfig	Configure and display network interface information	ifconfig
curl	Transfer data to or from a server. Usually used for web browsing	curl http://example.com
wget	Downloading files from the internet	wget http://example.com/file.txt

EXECUTE MULTIPLE COMMANDS

- **Semicolon (;)**
runs the next command regardless of whether the previous command succeeds or fails
pwd ; ls
- **Double ampersand characters (&&)**
runs the next command only if the previous command succeeds
mkdir test && cd test
- **Pipes (|)**
passes the output of the first command as input to the second command
ls /etc | grep hosts

CONNECTION - SSH AND NC

- Secure Shell (SSH) command is used to securely connect to a remote server or system over a network. It provides encrypted communication for executing commands on remote machines.

Example :

```
ssh kali@192.168.1.10
```

```
ssh kali@192.168.1.10 -p 2222
```

- The nc command, also known as netcat, is a versatile tool for working with network connections. It can be used for tasks like port scanning, file transfers, and creating network connections.

Example :

```
nc 192.168.1.10 4444
```

WEB SERVER

- Allows us to host a web application
- Common web server - apache2 and python

- Installing

```
sudo apt install apache2
```

```
sudo systemctl start apache2
```

```
sudo systemctl status apache2
```

- Default file location : /var/www/html
- Check the website at <http://localhost>

GETTING HELP

Sometimes we stumble across tools or command that we are not familiar or we forgot the usage.

There are two common ways to get help :

- `man <tool>`
- `<tool> --help`

```
curl(1)                                curl Manual                                curl(1)
NAME
    curl - transfer a URL

SYNOPSIS
    curl [options / URLs]

DESCRIPTION
    curl is a tool for transferring data from or to a server using URLs. It supports these protocols: DICT, FILE,
    FTP, FTPS, GOPHER, GOPHERS, HTTP, HTTPS, IMAP, IMAPS, LDAP, LDAPS, MQTT, POP3, POP3S, RTMP, RTMPS, RTSP, SCP,
    SFTP, SMB, SMBS, SMTP, SMTPS, TELNET, TFTP, WS and WSS.

    curl is powered by libcurl for all transfer-related features. See libcurl\(3\) for details.

URL
    The URL syntax is protocol-dependent. You find a detailed description in RFC 3986.

    If you provide a URL without a leading protocol:// scheme, curl guesses what protocol you want. It then de-
```

using man

```
coreteam@htb-vps-training:~$ curl --help
Usage: curl [options...] <url>
  -d, --data <data>           HTTP POST data
  -f, --fail                   Fail fast with no output on HTTP errors
  -h, --help <category>      Get help for commands
  -i, --include                Include protocol response headers in the output
  -o, --output <file>         Write to file instead of stdout
  -O, --remote-name            Write output to a file named as the remote file
  -s, --silent                 Silent mode
  -T, --upload-file <file>    Transfer local FILE to destination
  -u, --user <user:password>  Server user and password
  -A, --user-agent <name>     Send User-Agent <name> to server
  -v, --verbose                Make the operation more talkative
  -V, --version                Show version number and quit
```

This is not the full help, this menu is stripped into categories.
Use "--help category" to get an overview of all categories.
For all options use the manual or "--help all".

using --help



INTRO TO CTF

WHAT IS CAPTURE THE FLAG (CTF)?

Capture the Flag (CTF) in computer security is an exercise in which participants attempt to find text strings, called “flags”, which are secretly hidden in purposefully-vulnerable programs.

example:

```
ICECTF{Th1s_1s_4_sEcR3T_fLaG}
```

CTF FORMATS

- **Attack & Defence**

- Each team receives identical VM with vulnerabilities waiting to be exploited.
- Infiltrate other team's machine while protecting your own machine from being invaded

- **King of The Hill (KOTH)**

- Teams compete to control a single server or machine (hill) as long as possible
- Like Domination in CODM but only one point which is the "hill"

CTF FORMATS

- **Jeopardy**

- 90% of the CTF are held in Jeopardy format
- A series of cybersecurity challenge across various categories
- Solve challenge to earn the flag



CATEGORIES OF JEOPARDY CTF

1. **Forensics**
2. **Steganography** (can be categorised under Forensics)
3. **Web Exploitation**
4. **Cryptography**
5. **Reverse Engineering**
6. **Binary Exploitation / PWN**
7. **OSINT** (Open-Source Intelligence)
8. **Miscellaneous**



FORENSICS

- Forensics involves investigating **digital artifacts** to uncover evidence.
- Tasks might include analyzing **disk images**, examining **log files**, or **recovering deleted files** to find flags hidden in digital traces.



FORENSICS TOOLS

Wireshark

- The undisputed king of network analysis. Wireshark is a tool to capture and analyze network traffic. CTF players use it to identify suspicious activity and hidden messages within network packets.

Volatility / Volatility3

- A memory forensics utility framework to extract digital artifacts from volatile memory (RAM) samples.
- It is an open-source command-line tool that CTF players use to analyse RAM dumps

FORENSICS TOOLS

Autopsy

- Autopsy is a free digital forensics platform with a graphical interface to utilise **The Sleuthkit Tools**. It is often used by CTF players to analyse the contents of **disk images or memory dumps** and recover deleted files.

FTK Imager

- A free data preview and imaging tool that is primarily used to analyse the contents of forensic images or memory dumps and **mount the challenge image files as a drive on your PC**.

AccessData FTK Imager 3.14.6

File View Mode Help

Evidence Tree

- \\PHYSICALDRIVE0
 - Partition 1 [500649MB]
 - Partition 2 [99MB]
 - Partition 3 [202409MB]
 - NONAME [NTFS]
 - [orphan]
 - [root]
 - \$BadClus
 - \$Extend
 - \$Recycle.Bin
 - \$Secure
 - 3590F75ABA9E485486C100C1A9D4FF06ZZ.ZZZ.ZZZZZ
 - 3590F75ABA9E485486C100C1A9D4FF06ZZ.ZZZ.ZZZZZZ
 - 3590F75ABA9E485486C100C1A9D4FF06ZZ.ZZZ.ZZZ
 - 3590F75ABA9E485486C100C1A9D4FF06ZZ.ZZZ.ZZZZZ
 - 3590F75ABA9E485486C100C1A9D4FF06ZZ.ZZZ.ZZZZZ
 - 3590F75ABA9E485486C100C1A9D4FF06ZZ.ZZZ.ZZZZZ
 - 3590F75ABA9E485486C100C1A9D4FF06ZZ.ZZZ.ZZZ.ZZZ
 - Config.Msi
 - Defl
 - Documents and Settings
 - FFOutput

Evidence Tree Pane

File List

Name	Size	Type	Date Modified
\$Boot	8	Regular File	27-11-2012 8:5...
\$I30	13	NTFS Index All...	20-2-2014 7:16...
\$LogFile		File	27-11-2012 8:5...
\$MFT		File	27-11-2012 8:5...
\$MFTMirr	4	Regular File	27-11-2012 8:5...
\$Secure	1	Regular File	27-11-2012 8:5...

File List Pane

Viewer Pane

Hex Value Interpreter

Type	Size	Value
signed integer	1-8	
unsigned integer	1-8	
FILETIME (UTC)	8	
FILETIME (local)	8	
DOS date	2	
DOS time	2	
time_t (UTC)	4	
time_t (local)	4	

Hex Value Interpreter

Byte order: ☒ Little endian ☐ Big endian

Cursor pos = 0; clus = 786432; log sec = 6291456; phy sec = 1031829504

STEGANOGRAPHY

- Steganography is the practice of **hiding information within other non-secret data**, such as images or audio files.
- Participants need to identify and **extract hidden messages or files** that are concealed within seemingly innocuous files.

5wk6pk7Vd8rJwXfxPC3IDVewWJ2k3U
13OBxkCUHvhdvjAobtg3l84RFWF2Ph
xobC3QMcuqRmE51jKGCUaz0VJ9AnO
P9lg4RNz1OJZYgxIhzmNiA1OfxYgH2
Nl1sTb1mCwYbpJaXTk7xtUcDqmi5w
WeFXKxQ8qQRre1O7pGCUC0Ne1j6TG
juMzREWewsKxk7Vd0rI1sYdK2k7pHy
iqdEASjK2tTwqiH84PhJvXSa62Pg3Zq
WeTwOB2k7UcviASj6oTbYg3lErnFXf
3Q8Vivi6kIbBocZMzR9WnFXK3QMbi
bImnYbpJaXTkO7xtUcDqmi5wsfGOA
x7Vd8rJwXfxPm73jnN3YFjswlsa21si
Wqbig3l84RFWF2PuMznas21OhdI9Hm

STEGA TOOLS

Aperi'Solve

- A free and online platform which performs **layer analysis on images**. The platform integrates various steganography analysis tools like **zsteg, steghide, outguess, exiftool, binwalk, foremost, and strings** seamlessly and uses them all automatically on the picture to analyse it once it is uploaded.

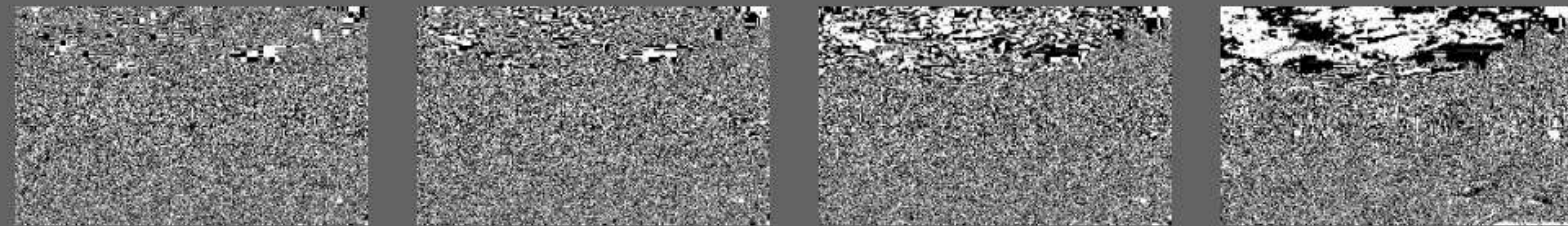
Audacity

- A free and open-source digital **audio editor** and recording application software, available for Windows, macOS, Linux, and other Unix-like operating systems.
- **Alternative – Wavacity**

Aperi'Solve



[+] Blue



Zsteg

```
imagedata .. text: "\n\n\r\r\r\r\r\r\r\r\r\r"
b1,b,msb,xy .. file: Apple DiskCopy 4.2 image \241\363\{jj9\241\371\307, 3874881423 bytes, 0xa165f370
tag size, 0xbc encoding, 0xcc format
b1,rgb,lsb,xy .. text: "tribectf{h1dd3n_1n_th3_bricks}"
b3,g,lsb,xy .. text: "?B5(j~I$"
b3,b,lsb,xy .. text: "JI&rImr{#"
b4,r,lsb,xy .. text: "wvfUDDDD\"32"
b4,g,lsb,xy .. text: "2#2U3#\\"
b4,b,lsb,xy .. text: "wgvvEE#2#!"
b4,rgb,lsb,xy .. text: "WUuFDdFDdFDdFDd%2S6Cd6BT"
b4,rgb,msb,xy .. text: "b\"&b\"&b\"&b\"&"
b4,bgr,lsb,xy .. text: "WUuFDdFDdFDdFDd5#RF4cF4R4"
b4,bgr,msb,xy .. text: "b\"&b\"&b\"&b\"&"
```

Steghide

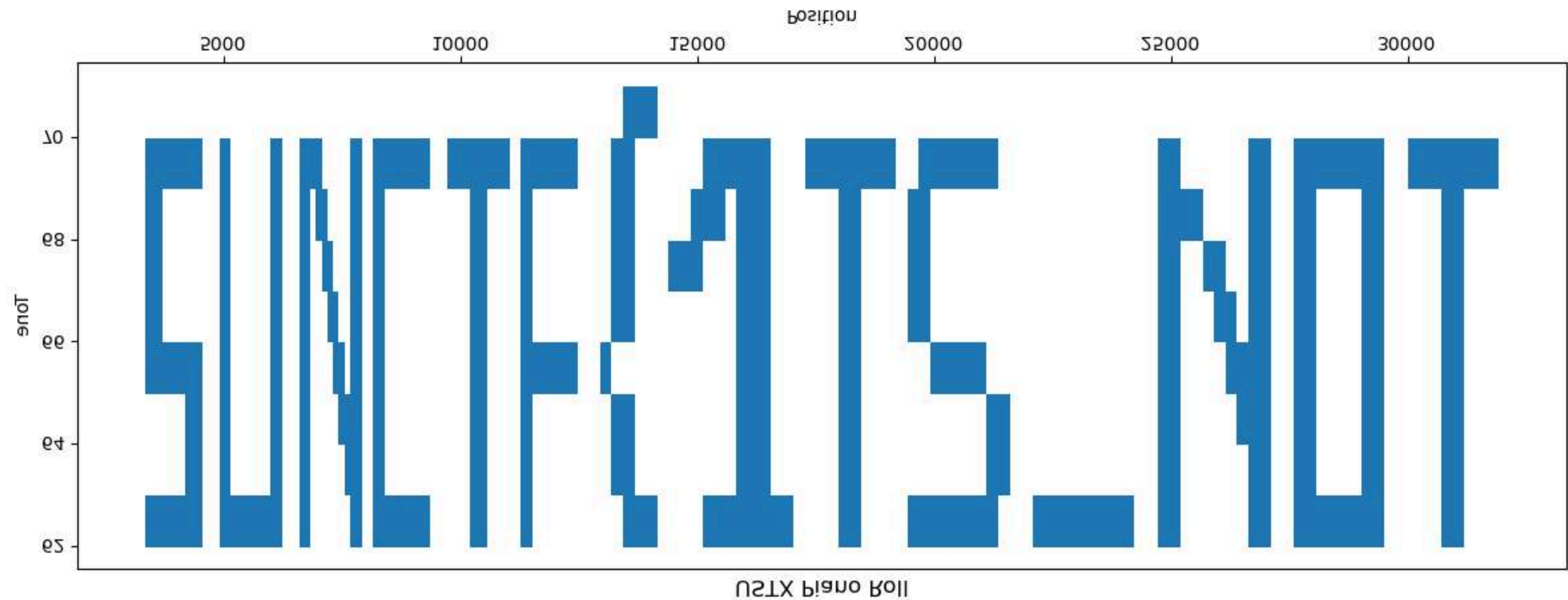
```
steghide: the file format of the file "/app/uploads/6c1e6d326588972ac1e6419853784ed6/image.png" is
not supported.
```

[DOWNLOAD FILES](#)

Outguess

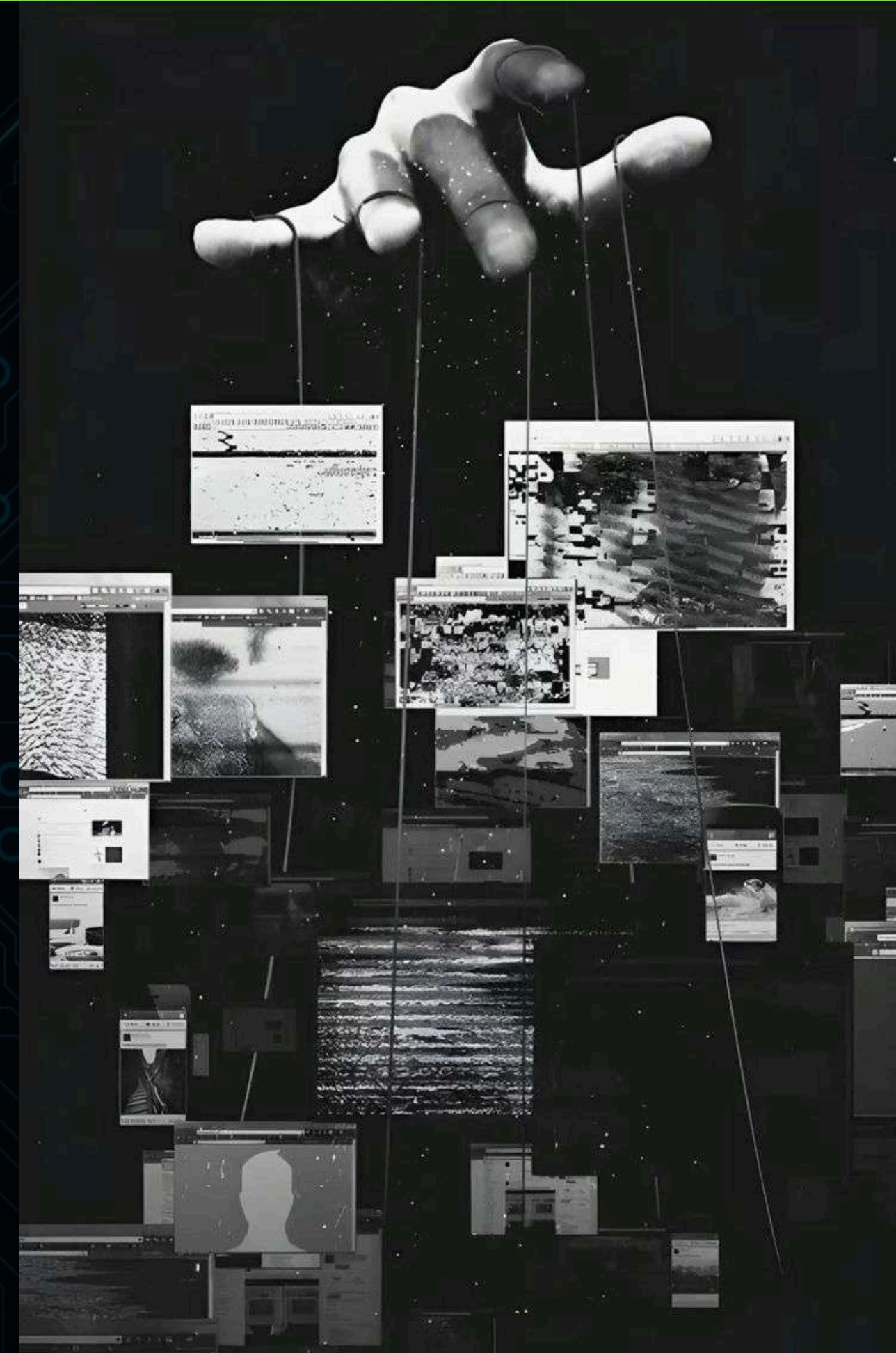
Unknown data type of /app/uploads/6c1e6d326588972ac1e6419853784ed6/image.png

Audio Steganography



WEB EXPLOITATION

- Web exploitation involves identifying and exploiting vulnerabilities in web applications.
- Participants may work with SQL injection, Cross-Site Scripting (XSS), or other web-based attacks to gain access to sensitive information or retrieve flags.



WEB TOOLS

Web Browser (Developer Tools).

- Players use the developer tools in the browsers to check the website **source files, cookies and HTTP requests/responses** going in and out of the browsers.
- Instructions to Access Developer Tools : **Ctrl+Shift+I / F12**

Burp Suite

- Very popular as an **interceptor and proxy server**. Use it to intercept communications between their browser and the web server to further examine and **modify requests and responses** before sending them.

WEB TOOLS

Gobuster

- Written in Go, Gobuster is a champion of speed when it comes to **brute-forcing directories on websites**, (Can also bruteforce files, DNS subdomains, Virtual Host names, Amazon S3 buckets, Google Cloud buckets, and TFTP servers)

Gobuster

```
root@n3n0sd0n4ld:~/CyberSploit# gobuster dir -u http://cybersploit  
-w /usr/share/dirb/wordlists/common.txt
```

Gobuster v3.0.1

by OJ Reeves (@TheColonial) & Christian Mehlmauer (@_FireFart_)

[+] Url: http://cybersploit
[+] Threads: 10
[+] Wordlist: /usr/share/dirb/wordlists/common.txt
[+] Status codes: 200,204,301,302,307,401,403
[+] User Agent: gobuster/3.0.1
[+] Timeout: 10s

2020/07/10 17:14:58 Starting gobuster

/.htpasswd (Status: 403)
/.htaccess (Status: 403)
/.hta (Status: 403)
/cgi-bin/ (Status: 403)
/index.html (Status: 200)
/index (Status: 200)
/hacker (Status: 200)
/robots.txt (Status: 200) ←
/robots (Status: 200)
/server-status (Status: 403)

CRYPTOGRAPHY

- This category involves solving puzzles related to **encryption and decryption**.
- Participants might be required to **crack ciphers, decode messages, or analyze cryptographic protocols** to retrieve hidden flags.



CRYPTO TOOLS

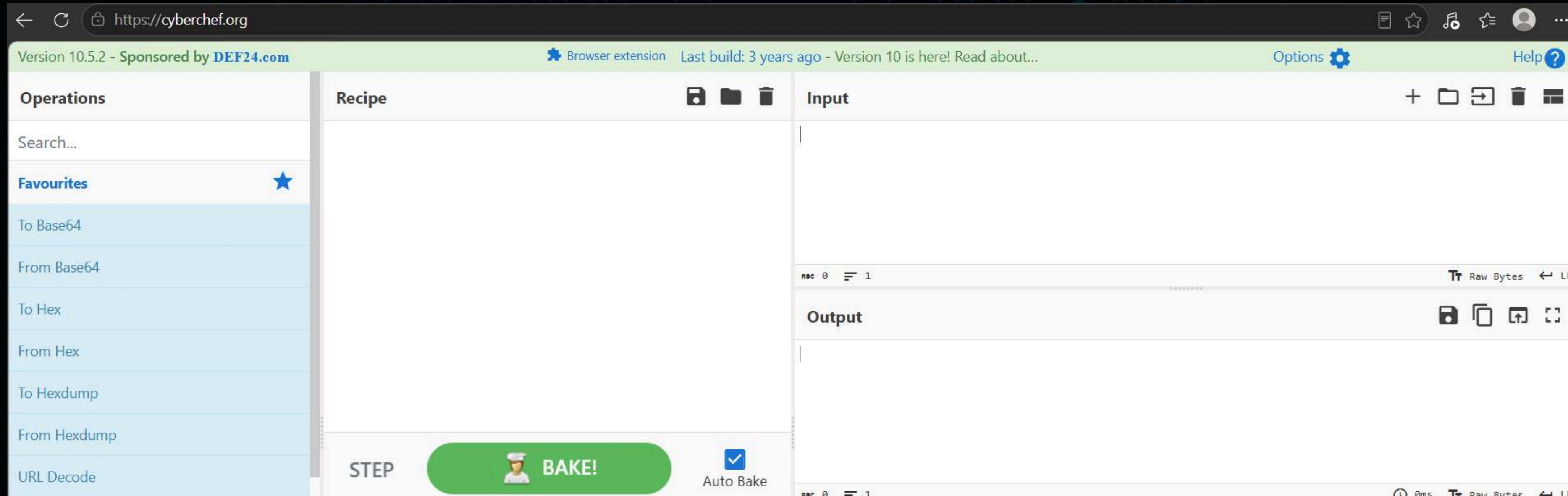
CyberChef

- The Swiss Army Knife of Crypto. CyberChef offers a wide range of functionalities, from **encoding and decoding** messages to **cracking simple ciphers**.

dCode

- Dcode is a treasure trove of online tools. It supports a **vast array of ciphers**, from classic Caesar ciphers to more complex ones.
- Tips – you can use **Cipher Identifier to detect cipher used**, works like a charm... sometimes

Cyberchef – <https://cyberchef.org/>



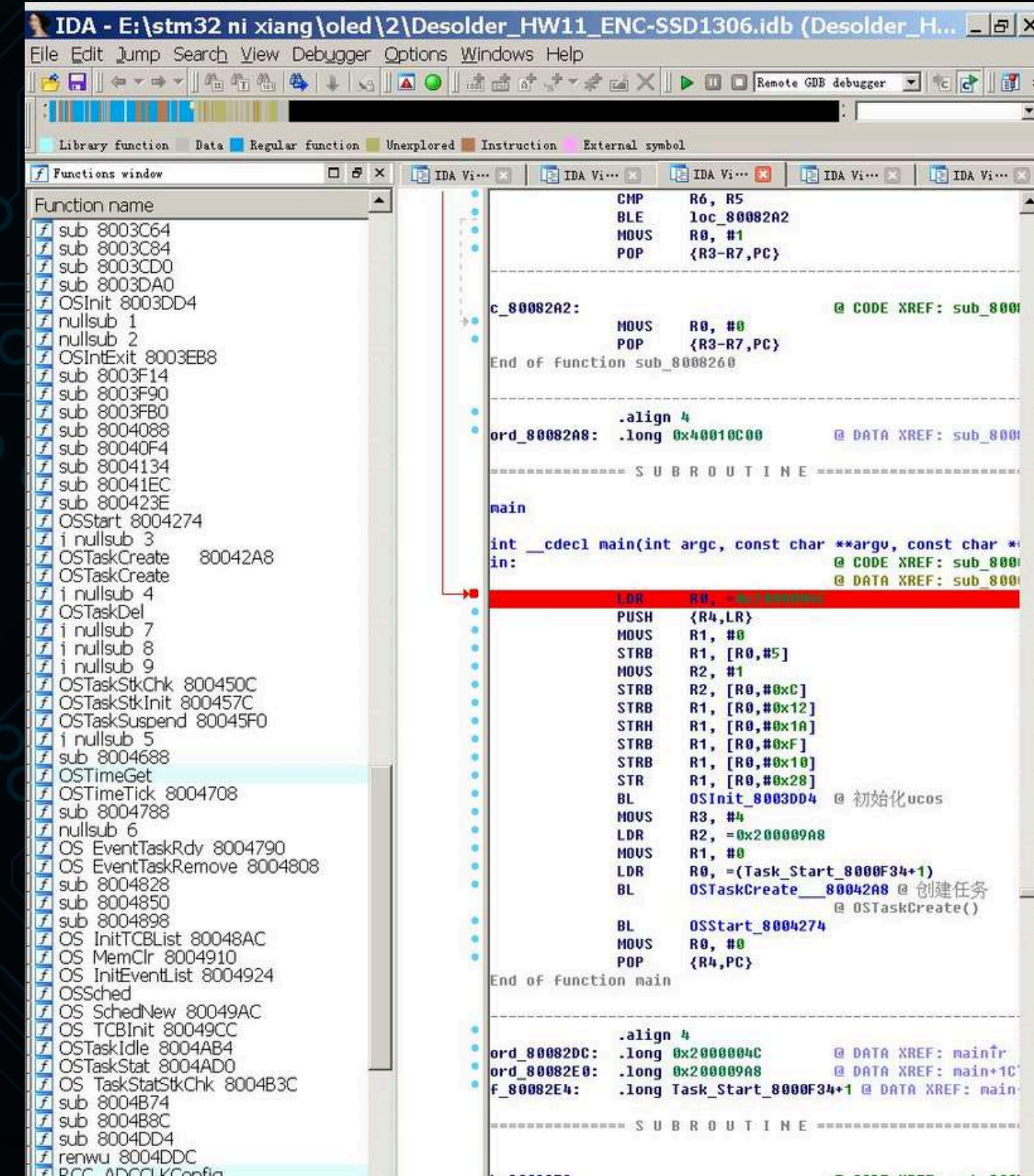
CRYPTO TOOLS

CrackStation – for hash cracking

- Uses **precomputed tables** to look up hashes and recover matching **plaintext passwords**.
- Useful when you find a password hash (e.g., MD5 or SHA-1) and need to identify the original password.
- Works best for **common passwords**; it generally cannot crack strong, unique passwords that aren't in its database.
- Other similar tool is **hashes.com**

REVERSE ENGINEERING

- This category involves **dissecting software** or binaries to understand their **functionality** and identify vulnerabilities.
- Participants might analyze executable files to determine **how they work**, **identify flaws**, or **recover source code** to find flags.
- The tools has 2 categories:
Debugger
Disassembler



RE TOOLS

Feature	Disassembler	Debugger
Type of Analysis	Static (analyzes code without running it)	Dynamic (runs the code and observes behavior)
Inspection	Examines code structure, control flow, and data	Inspects live memory, registers, and execution
Usage	Understands how code works, sees hidden logic	Observes, modifies, or manipulates the program during execution
Modification	Limited; only changes are theoretical (patches)	Allows real-time modification of code and memory
Tools	IDA Pro, Ghidra, Radare2, Binary Ninja	GDB, x64dbg, WinDbg, OllyDbg

RE TOOLS

IDA

- Short for Interactive Disassembler, IDA. It is able to **extract** a lot of **information from binary files** which includes but are not limited to the graph flows, strings, exports, and imports.

Ghidra

- A free and open-source alternative to commercial tools like IDA, Developed by the National Security Agency (NSA)
- Both tools can be used as a **disassembler, debugger and decompiler.**

RE TOOLS

GDB

- The GNU Debugger, or just GDB is a powerful **command line debugger** used to execute a compiled program step-by-step. Allows players to read the **program's assembly code**, **inspect register values** and understand what happens when it crashes.

DogBolt

- An online tools that contains some of the popular RE tools like, Ghidra, Binary Ninja and angr.
- It can automatically **analyze and decompile** binary files, **converting them into more readable code**.

Dogbolt – <https://dogbolt.org/>



Decompiler Explorer

[What is this?](#)

Upload File

Your file must be **less than 2MB** in size. Uploaded binaries [are retained](#).

Scegli file Nessun file selezionato

Samples

Or check out one of these samples we've provided:

Hello World on Arm Linux

☒ angr

☒ BinaryNinja

☐ Boomerang

☐ dewolf

☒ Ghidra

☒ Hex-Rays

☐ RecStudio

☐ Reko

☐ Relyze

☐ RetDec

☒ Snowman

angr

9.2.79

```
1 int _init()
2 {
3     return (unsigned int)call_weak_fn();
4 }
5
6 int sub_10308()
7 {
8     unsigned int v0; // [bp-0x4]
9     unsigned int v1; // lr
10
11     v0 = v1;
12     goto *((int*)(66328 + 68840 + 8));
13 }
14
15 int _start(unsigned int a0, unsigned int a1, unsigned int a2)
16 {
17     unsigned int v0; // [bp-0x4]
18
19     v0 = a0;
20     __libc_start_main(main, a4, &a4, NULL, NULL,
21     21 }
22
23 int sub_10390()
24 {
25     abort(); /* do not return */
26 }
27
28 void call_weak_fn()
29 {
30     if (!(!0))
31         goto LABEL_103b4;
32     return;
33 LABEL_103b4:
34     __gmon_start__();
35     return;
36 }
37
38 extern char TMC_FND :
```

BinaryNinja

3.5.4526 (ec37d737)

```
1 int32_t _init(int32_t arg1, int32_t arg2)
2 {
3     return call_weak_fn(arg1, arg2);
4 }
5
6 int32_t sub_10308()
7 {
8     /* jump -> 0 */
9 }
10
11 void __libc_start_main(int32_t (* main)(int32_t
12 {
13     /* tailcall */
14     return __libc_start_main(main, argc, ubp_av.
15 }
16
17 int32_t getchar()
18 {
19     /* tailcall */
20     return getchar();
21 }
22
23 int32_t puts(char const* str)
24 {
25     /* tailcall */
26     return puts(str);
27 }
28
29 int32_t __gmon_start__()
30 {
31     /* tailcall */
32     return __gmon_start__();
33 }
34
35 void abort() __noreturn
36 {
37     /* tailcall */
38     return abort();
39 }
```

Ghidra

10.4 (1801dc1e)

```
1 #include "out.h"
2
3
4
5 int _init(EVP_PKEY_CTX *ctx)
6 {
7     {
8         int iVar1;
9
10        iVar1 = call_weak_fn();
11        return iVar1;
12    }
13
14
15 void __libc_start_main(void)
16 {
17     {
18         __libc_start_main();
19         return;
20     }
21 }
22
23 // WARNING: Unknown calling convention -- yet no
24
25 int getchar(void)
26 {
27     {
28         int iVar1;
29
30         iVar1 = getchar();
31         return iVar1;
32     }
33 }
34
35 // WARNING: Unknown calling convention -- yet no
```

Hex-Rays

8.3.0.230608

```
1 /* This file was generated by the Hex-Rays decompiler.
2 Copyright (c) 2007-2021 Hex-Rays <info@hex-rays.com>
3
4 Detected compiler: GNU C++
5 */
6
7 #include <defs.h>
8
9 #include <stdarg.h>
10
11 //-----
12 // Function declarations
13
14 int init_proc();
15 void sub_10308();
16 // int __fastcall __libc_start_main(int (__fastcall
17 // int getchar(void);
18 // int puts(const char *s);
19 // int __gmon_start__(void); weak
20 // void abort(void);
21 void __noreturn start(void (*a1)(void), int a2,
22 int call_weak_fn();
23 char *deregister_tm_clones();
24 __int64 register_tm_clones();
25 char *_do_global_dtors_aux();
26 int __fastcall main(int argc, const char **argv,
27 void term_proc();
28
29 //-----
30 // Data declarations
31
32 Elf32_Dyn **off_10394 = (Elf32_Dyn **)0x10C6C;
33 char _bss_start; // weak
34 // extern _UNKNOWN __gmon_start__; weak
35
36 //----- (000102FC) -----
```

Snowman

0.1.2-21 (g6fed71c)

```
125 int1_t z3;
126 int1_t z4;
127 int32_t r0_5;
128 int1_t z6;
129 int32_t r3_7;
130
131 if (z2) {
132 }
133 if (z3) {
134 }
135 if (z4) {
136 }
137 r0_5 = d;
138 if (z6) {
139     return r0_5;
140 } else {
141     r3_7 = g10420;
142     if (!r3_7) {
143         return r0_5;
144     } else {
145         goto r3_7;
146     }
147 }
148 }
149
150 /* $d */
151 unsigned char* d = reinterpret_cast<unsigned char*>(
152
153 /* $d */
154 uint32_t d(uint32_t r0) {
155     int1_t z2;
156     int1_t z3;
157     int1_t z4;
158     unsigned char* r4_5;
159     uint32_t r0_6;
160
161     if (z2) {
162     }
163 }
```

BINARY EXPLOITATION/PWN

- This category focuses on exploiting vulnerabilities in compiled binary applications.
- Tasks typically involve techniques like **buffer overflow** attacks or **memory corruption** to gain control of a program and access hidden flags.

A	01000001	
B	01000010	010000010
C	01000011	
D	01000100	B 01001001
E	01000101	I 01001110
F	01000110	N 01000001
G	01000111	A 01010010
H	01001000	R 01011001
I	01001001	
J	01001010	
K	01001011	
L	01001100	
M	01001101	
N	01001110	
O	01001111	
P	01010000	
Q	01010001	
R	01010010	C 01000011
S	01010011	
T	01010100	O 01001111
U	01010101	
V	01010110	D 01000100
W	01010111	
X	01011000	E 01000101
Y	01011001	
Z	01011010	

PWN TOOLS

- **pwntools**
 - A CTF framework and **exploit development library** written in Python. Basically, it contains many tools that make exploit writing in CTFs as simple as possible and at the same time, easy to read.
- **checksec**
 - A **bash script** used by CTF players to **scan and test binaries** for standard security features used in Linux operating systems. The security features that can be revealed include Stack Canaries, NX (No eXecute) protection, RELRO, and Position Independent Executables (PIE).

PWN TOOLS

ROPgadget

- Using the Capstone disassembler. The ROPgadget tool is used to search for **gadgets** (Assembly segments that perform the operation of popping one or more registers from the stack. followed by a ret instruction) on binaries to facilitate ROP exploitation.
- It is able to support binaries in the ELF/PE/Mach-O/Raw formats on x86, x64, ARM, ARM64, PowerPC, SPARC, MIPS, RISC-V 64, and RISC-V Compressed architectures.

OSINT (OPEN-SOURCE INTELLIGENCE)

- This category focuses on gathering information from publicly available sources.
- Participants might analyze social media, websites, public records, and other accessible data to find clues or flags related to a specific topic or target.



OSINT TOOLS

Google Images

- A powerful reverse image search function that allows you to **upload the image or provide its URL** and see where else it appears on the internet.
- This can lead you to websites containing **valuable clues** or the image itself.

WayBack Machine

- A service that allows its users to **go "back in time"** to view **snapshots of web pages from different dates**. It is basically a digital archive of the World Wide Web founded by the Internet Archive.

Example challenge style i've encountered before....

HIGHWAY I

50

👍 8 (88% liked) 👎 1

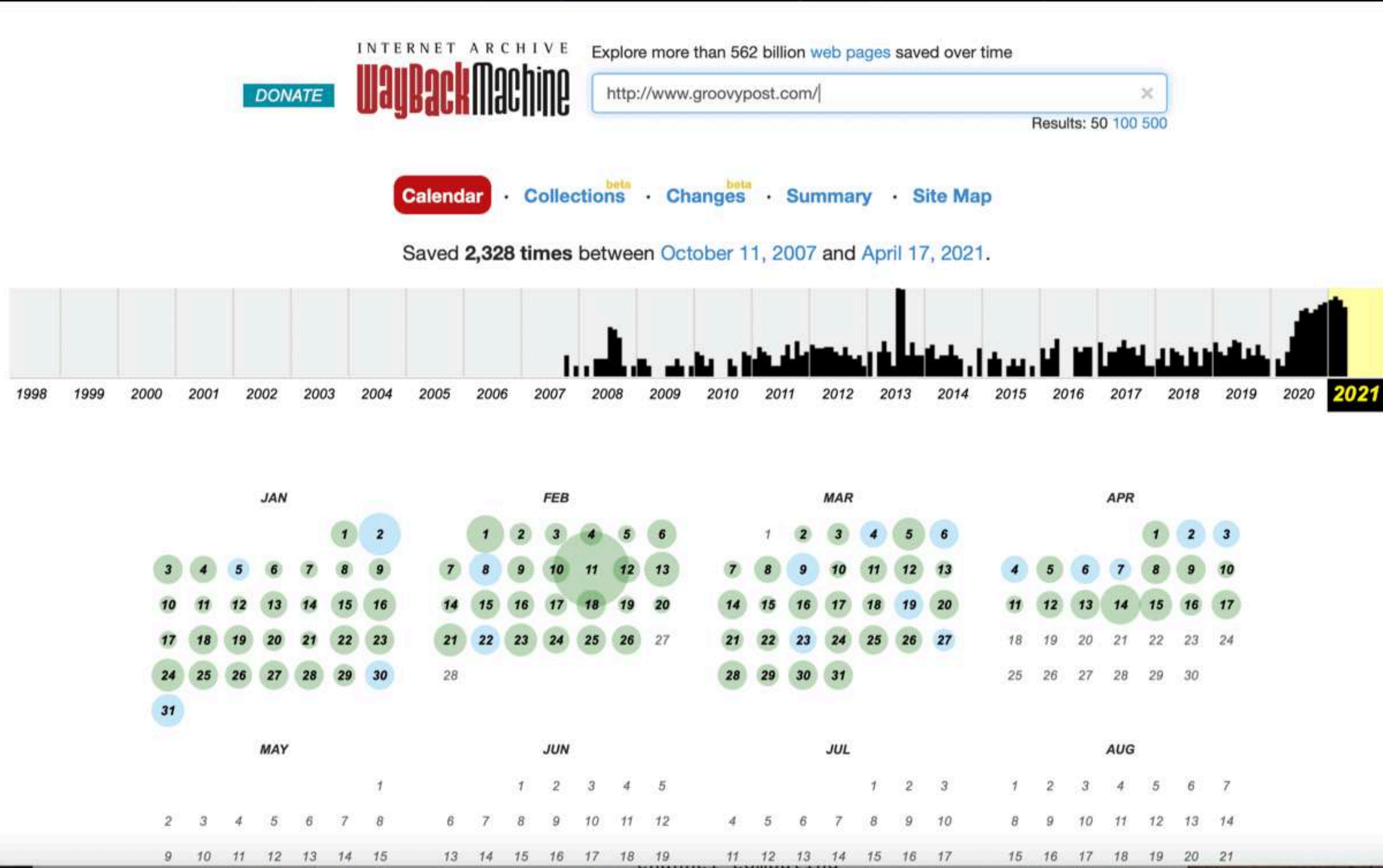
Can you find this highway?

Submit in 3 decimal places

Flag Format: HNYX{latitude,longitude}



Wayback Machine – <https://web.archive.org/>



OSINT TOOLS

Social Media Accounts

- Popular social media platforms like LinkedIn and X with millions of active users are breeding grounds for **publicly available information**, making them ideal hunting grounds for OSINT challenges.
- Commonly Used Platform in OSINT challenges:
 - X (Twitter)
 - Telegram
 - Github
 - Tiktok
 - LinkedIn

MISCELLANEOUS

- This category encompasses challenges that **don't fit neatly into other categories**.
- There are **no specific tools** designated for this category. Participants may need to use general cybersecurity tools, research strategies, or even creative problem-solving techniques depending on the nature of the challenge.



AND... MORE CATEGORIES!

AI Security

**Hardware
Hacking**

**Post-Quantum
Crypto (PQC)**

**Boot-to-Root
(B2R)**

Cloud Security

DFIR

**Malware
Analysis**

**Advanced
Persistent Threat
(APT)**

Mobile Security

NEW TO CTF? CHECK THIS OUT!

- Basic Linux CLI – <https://picoctfsolutions.com/posts/linux-cli-ctf>
- CTF Primer – <https://primer.cylabacademy.org/>
- Solo Practice – <https://learn.cylabacademy.org/library?page=1>
- Practice with you team – <https://skrctf.me/>



HANDS-ON CHALLENGE



THANK YOU